



10th Disease Maps Community Meeting

PROGRAM & ABSTRACT BOOKLET

5-7 October 2026

University of Rostock · Main Building · Universitätsplatz 1 · 18055 Rostock · Germany

Welcome to DMCM2026

Welcome to the 10th Disease Maps Community Meeting in Rostock. Over three days, the program brings together researchers who construct, curate, analyze and apply mechanistic representations of disease and biological processes. The meeting combines invited keynotes, contributed talks, poster sessions, public discussion rounds and community breakouts, with particular emphasis on computational disease modeling, AI-assisted knowledge curation, systems biology and clinical translation.

DMCM2026 is organized by the Department of Systems Biology and Bioinformatics at the University of Rostock. Participation is free of charge with registration, reflecting the open and community-driven character of the meeting. The tenth edition is also an opportunity to look back at a decade of Disease Maps Community Meetings and to discuss the standards, infrastructures and collaborations needed for the next phase of the field.

Conference at a glance

Dates 5-7 October 2026

Venue Historic Main Building, University of Rostock

Address Universitätsplatz 1, 18055 Rostock, Germany

Monday / Tuesday Aula, Main Building

Wednesday Lecture Hall 03 (HS 03)

Website <https://disease-maps.io/DMCM2026/>

Organizing team Matti van Welzen · Alexandra Baumann · Shailendra Gupta · Olaf Wolkenhauer · Marek Ostaszewski

Practical information

Venue and arrival

University of Rostock, Main Building, Universitätsplatz 1, 18055 Rostock, Germany. The building is centrally located. From Rostock Central Station, tram services connect to the city center; the final walk to Universitätsplatz is only a few minutes.

Rooms

Monday and Tuesday: Aula, Main Building. Wednesday: Lecture Hall 03 (HS 03).

Travel

Rostock can be reached by direct train connections from Hamburg and Berlin. Local and regional trains also connect Rostock with Warnemünde and the Baltic coast.

Conference website

<https://disease-maps.io/DMCM2026/>

Further information

- Disease Maps Community: <https://disease-maps.io/about/>
- Disease Maps meeting history: <https://disease-maps.io/events/>
- University of Rostock: <https://www.uni-rostock.de/>
- Rostock city information: <https://www.rostock.de/>

About the Disease Maps Community

The Disease Maps Project is a large-scale community effort initiated by the Luxembourg Centre for Systems Biomedicine, the European Institute for Systems Biology and Medicine and Institut Curie. Its central idea is to organize mechanistic knowledge about diseases and biological systems into computable, interoperable molecular maps that can serve as curated knowledge resources, interfaces for data interpretation and starting points for modeling.

Disease Maps connect expertise distributed across experimental and clinical research, literature curation, pathway representation, software engineering, standards, data integration and mathematical modeling. By encoding mechanisms in structured formats, maps can support visual exploration, network analysis, omics overlays, logical or mechanistic modeling and the formulation of testable hypotheses.

A decade of Disease Maps Community Meetings

1st · 15-17 Feb 2017	Lyon, France · Best practices, map quality and translational applications
2nd · 2-3 Oct 2017	Belval, Luxembourg · Linked to the International Systems Biomedicine Symposium
3rd · 21-22 Jun 2018	Paris, France · Resources, standards, modeling and hands-on tutorials
4th · 2-4 Oct 2019	Sevilla, Spain · Disease-map construction and modeling applications
5th · 12-14 Nov 2020	Online · Strong focus on the COVID-19 Disease Map
6th · 29-30 Nov 2021	Online · Organized by the University of Rostock SBI department
7th · 3-5 Apr 2023	Maastricht, Netherlands · "The Disease Maps Alliance: Bringing Experts Together"
8th · 25-27 Mar 2024	Belval, Luxembourg · Joint event with the ELIXIR Systems Biology Community
9th · 15-17 Apr 2025	Leuven, Belgium · Hosted at KU Leuven with the University of Liège
10th · 5-7 Oct 2026	Rostock, Germany · Tenth-meeting milestone hosted by the University of Rostock

In memoriam · Prof. Reinhard Schneider

DMCM2026 remembers Prof. Reinhard Schneider, a long-standing leader in computational biology and a founding principal investigator of the Disease Maps effort. At the Luxembourg Centre for Systems Biomedicine, he led the Bioinformatics Core and contributed expertise spanning systems biology, high-performance computing, data integration, visualization, text mining and knowledge management.

Reinhard was closely connected to the development of the Disease Maps community from its early years, appearing repeatedly in the meeting series as organizer, coordinator and speaker. At this tenth community meeting we acknowledge his role in building the scientific and personal networks that allowed Disease Maps to grow into an international collaborative effort.

The University of Rostock

Founded in 1419, the University of Rostock is the oldest university in the Baltic Sea region and one of the oldest universities in Northern Europe. DMCM2026 takes place in the historic Main Building on Universitätsplatz. The Aula provides the setting for the first two conference days; Wednesday's breakout program moves to Lecture Hall 03 (HS 03).

The Department of Systems Biology and Bioinformatics organizes DMCM2026 and works at the interface of mathematical modeling, data analysis and molecular medicine. Rostock has contributed to earlier Disease Maps meetings and projects, making the tenth meeting a return to a community long connected to the university.

Rostock · Hanseatic city by the Baltic Sea

Rostock is a Hanseatic and university city on the River Warnow, close to the Baltic Sea. Its historic center reflects its maritime past through brick Gothic architecture, churches, market squares and surviving sections of the medieval city fortifications. The conference venue sits in the heart of the city center, with many central sights, restaurants and hotels within walking distance.

Rostock Central Station is connected to the center by tram, and the city can be reached by direct rail services from Hamburg and Berlin. Participants extending their stay can also reach Warnemünde and the Baltic coast easily by local public transport.

Sponsor · Heel GmbH

DMCM2026 is supported by Heel GmbH, an international pharmaceutical company based in Baden-Baden, Germany. Heel researchers are represented in contributions to the meeting, including work on MASLD and osteoarthritis pain-signaling disease maps. Website:

<https://www.heel.com/about-heel>

Sponsor presentation · Tuesday, 16:45-17:30

Myron Schultz, Heel GmbH — From Multicomponent Medicines to Network Pharmacology: Heel's Journey in Systems Research

Keynote speakers

SESSION 1 · COMPUTATIONAL MODELS OF DISEASE PROCESSES



Marek Ostaszewski

Luxembourg Centre for Systems Biomedicine (LCSB), University of Luxembourg

Disease Maps Community: bridging knowledge and models with standards, software, and suffering

Marek Ostaszewski is a senior researcher at the Luxembourg Centre for Systems Biomedicine, University of Luxembourg. His work focuses on systems-biology approaches to human disease, molecular network construction and analysis, visual exploration of biomedical knowledge and data, and translational applications in neurodegenerative and chronic immune disorders.

SESSION 2 · APPLICATIONS FOR AI IN MODEL CURATION, EXPLORATION, AND ANALYSIS · REMOTE



Sebastian Lobentanzer

Helmholtz Munich, Germany

Powerful ≠ Ready: AI and knowledge representation in biomedicine

Dr. Sebastian Lobentanzer is a Principal Investigator at Helmholtz Munich, a Helmholtz AI Young Investigator, and head of the Computational Biology Unit at the German Center for Diabetes Research. His research asks how scientific work can be automated reliably, with a focus on biomedical knowledge graphs, large language models, agentic AI and the governance and evaluation of AI systems for science.

SESSION 3 · RECENT ADVANCES IN SYSTEMS BIOLOGY



Olaf Wolkenhauer

University of Rostock, Germany

Disease Maps - Repositories or models?

Olaf Wolkenhauer received degrees in systems and control engineering in Hamburg and Portsmouth and obtained his PhD from the University of Manchester Institute of Science and Technology in 1997. He was appointed Professor at the University of Rostock in 2003. His research focuses on uncertainty in the analysis and interpretation of data in the life sciences.

SESSION 4 · SYSTEMS MEDICINE AND CLINICAL TRANSLATION



Markus Wolfien

Institute for Medical Informatics and Biometry, TUD Dresden University of Technology, Germany

From Maps to Decisions - Disease Maps as Mechanistic Infrastructure for Precision Medicine

Dr.-Ing. Markus Wolfien is Group Lead for Data Science & Bioinformatics and Deputy Head of the Chair for Medical Informatics at the Institute for Medical Informatics and Biometry, TU Dresden. His work spans translational systems medicine, biomedical data science, multi-omics, AI-based methods, interoperable clinical data infrastructures and mechanistic disease modeling for precision medicine and clinical decision support.

Monday, 5 October 2026

Room: Aula, Main Building

Time	Program
08:00 - 09:00	Breakfast, registration and arrival
09:00 - 09:30	Opening remarks, welcome to Rostock and remembrance of Prof. Reinhard Schneider
SESSION 1 · COMPUTATIONAL MODELS OF DISEASE PROCESSES Chair: David Safranek	
09:30 - 10:30	Keynote · Marek Ostaszewski, University of Luxembourg <i>Disease Maps Community: bridging knowledge and models with standards, software, and suffering</i>
10:30 - 11:00	Rupert Overall — EthoMap: A hybrid PD-AF knowledgebase of laboratory mouse behavior
11:00 - 11:15	Coffee break
11:15 - 11:45	Suman Muralidharan — AKOP: A Disease-Map Framework for Integrating Osteoarthritis Phenotypes, Molecular Networks and Transcriptomic Perturbations
11:45 - 12:15	Joshua Eugenio — Identifying Key Events for Nitrogen Dioxide Effects: Toward an Adverse Outcome Pathway
12:15 - 12:45	Shailendra Gupta — NOSE-OE: A Molecular Signaling Map of the Olfactory Epithelium
12:45 - 13:15	Matti van Welzen — Data Management and Analysis with Disease Maps in Interdisciplinary Research Groups
13:15 - 14:30	Lunch and poster viewing
SESSION 2 · APPLICATIONS FOR AI IN MODEL CURATION, EXPLORATION, AND ANALYSIS Chair: Marek Ostaszewski	
14:30 - 15:30	Keynote [remote] · Sebastian Lobentanzer, Helmholtz Center Munich, Germany <i>Powerful ≠ Ready: AI and knowledge representation in biomedicine</i>
15:30 - 16:00	Vaishnavi Sirul Velaga — Specialized Pro-Resolving Lipid Mediators (SPMs): Literature Analysis and an LLM-Driven Knowledge Graph
16:00 - 16:30	Joost Rehder — LLM-Assisted Graph Extraction for Disease Map Construction
16:30 - 16:45	Coffee break
16:45 - 18:00	Discussion round · How can AI support the construction, curation, and analysis of Disease Maps?
18:00 - 21:00	Dinner + socializing

Tuesday, 6 October 2026

Room: Aula, Main Building

Time	Program
08:00 - 09:00	Breakfast and arrival
SESSION 3 · RECENT ADVANCES IN SYSTEMS BIOLOGY Chair: Rupert Overall	
09:00 - 10:00	Keynote · Olaf Wolkenhauer, University of Rostock <i>Disease Maps - Repositories or models?</i>
10:00 - 10:30	Raul Armando Duran de Alba — Boolean network models recover the known biology of FSGS in podocytes and parietal epithelial cells
10:30 - 10:45	Coffee break
10:45 - 11:15	Hala Bouzidi — A Hybrid computational Framework Integrating Boolean Regulatory Networks and Agent-Based modalities for modeling the growth of colorectal cancer organoids
11:15 - 11:45	Tushar Patel — Enhancing the performance and interpretability of epigenetic clocks
11:45 - 12:15	Adrien Rougny — The momapy toolkit: Python tools for programmatic work with molecular maps
12:15 - 12:45	Abrar Zahin — Pathway Converter: A Desktop Workflow for Pathway Exploration and MINERVA-Based Conversion
12:45 - 14:00	Lunch and poster viewing
SESSION 4 · SYSTEMS MEDICINE AND CLINICAL TRANSLATION Chair: Adrien Rougny	
14:00 - 15:00	Keynote · Markus Wolfien, Institute for Medical Informatics and Biometry, Dresden <i>From Maps to Decisions - Disease Maps as Mechanistic Infrastructure for Precision Medicine</i>
15:00 - 15:30	Philip Stegmaier — Modeling of NF-κB Pathway Rewiring Reveals Clarithromycin for IBD
15:30 - 16:00	Geraldine Canny — Endometriosis-associated signaling pathways and potential for therapeutic development
16:00 - 16:30	Shailendra Gupta — A mechanistic pain signaling disease map enables pharmacophore-guided target prioritization and simulation of pain therapy response in osteoarthritis
16:30 - 16:45	Coffee break
16:45 - 17:30	Sponsor talk · Myron Schultz, Heel GmbH <i>From Multicomponent Medicines to Network Pharmacology: Heel's Journey in Systems Research</i>
17:30 - 18:30	Discussion round · How can we translate mechanistic disease models into clinical applications?
18:30 - 21:00	Dinner + socializing

Wednesday, 7 October 2026

Room: Lecture Hall 03 (HS 03)

Time	Program
08:00 - 09:00	Breakfast and arrival
09:00 - 09:30	Introduction to the community breakout round
09:30 - 10:30	Breakout rounds · working topics and collaborative roadmap development
10:30 - 11:00	Coffee break
11:00 - 12:00	Continuing breakout rounds
12:00 - 13:00	Self-organized lunch
13:00 - 14:00	Breakout summaries
14:00 - 14:30	Closing remarks

Poster presentations

Posters are available during the designated poster-viewing periods on Monday and Tuesday. Contributions submitted for both oral and poster presentation appear in both formats. Poster titles below reflect the final booklet titles.

P01 Boolean network models recover the known biology of FSGS in podocytes and parietal epithelial cells.

Raul Armando Duran De Alba, Olivia Lenoir, Pierre-Louis Tharaux, Clemence Reda

P02 A Hybrid computational Framework Integrating Boolean Regulatory Networks and Agent-Based modalities for modeling the growth of colorectal cancer organoids

Hala Bouzidi, Laurent Naudin, Philippe Castera, Guillaume Lefebvre

P03 Establishing an omics-informed logic-based quantitative adverse outcome pathway framework using PFOA-induced toxicity as a case study

Luiz Ladeira, Arthur de Carvalho e Silva, Bernard Staumont, Liesbet Geris

P04 Identifying Key Events for Nitrogen Dioxide Effects: Toward an Adverse Outcome Pathway

Joshua Eugenio, Véronique de Bruijn, Jeroen Pennings, Arno Swart, Pim Wassenaar, Aurélie Carlier, Yvonne Staal

P05 A disease agnostic pipeline for genome-scale metabolic modeling application in sample-specific drug combination ranking.

Rui Afonso Tavares; E. Stalidzans

P06 Toward explainable Disease Maps: Identifying best practices for designing modeling-ready diagrams

Pierre Klemmer, Samuel Pastva, David Šafránek, Marek Ostaszewski

P07 Epigenomic Instability - Is Epigenetic Age Acceleration a Survival Indicator in Lung Cancer?

Alexandra Baumann, Michael Seifert, Zholdas Buribayev, Olaf Wolkenhauer, Markus Wolfien

P08 A multi-layer disease map to study post-COVID mechanisms

Dries Heylen, Michelle Moerel, Martina Summer-Kutmon

P09 Specialized Pro-Resolving Lipid Mediators (SPMs): Literature Analysis and an LLM-Driven Knowledge Graph

Vaishnavi Sirul Velaga, Suman Muralidharan, Matti van Welzen, Konstantin Cesnulevicius, David Lescheid, Myron Schultz, Olaf Wolkenhauer, Shailendra Gupta

P10 A Multi-Compartment Disease Map Approach to MASLD

Matti van Welzen, Ronja Lea Jennifer Müller, Ali Canbay, Andreas Geier, Jörn M Schattenberg, Konstantin Cesnulevicius, Myron Schultz, David Lescheid, Olaf Wolkenhauer, Shailendra Gupta

P11 Network integration of liver-derived spatial transcriptomes from MASH patients

Lukas Erkan Neumann; Matti van Welzen; Shailendra Gupta (provisional author list)

Abstract index

ID	ABSTRACT & PRESENTER & FORMAT
A01	Boolean network models recover the known biology of FSGS in podocytes and parietal epithelial cells Raul Armando DURAN DE ALBA • Talk + Poster
A02	A Hybrid computational Framework Integrating Boolean Regulatory Networks and Agent-Based modalities for modeling the growth of colorectal cancer organoids Hala Bouzidi • Talk + Poster
A03	Establishing an omics-informed logic-based quantitative adverse outcome pathway framework using PFOA-induced toxicity as a case study Luiz Ladeira • Poster
A04	Enhancing the performance and interpretability of epigenetic clocks Tushar Patel • Talk
A05	Identifying Key Events for Nitrogen Dioxide Effects: Toward an Adverse Outcome Pathway Joshua Eugenio • Talk + Poster
A06	LLM-Assisted Graph Extraction for Disease Map Construction Joost Rehder • Talk
A07	The momapy toolkit: Python tools for programmatic work with molecular maps Adrien Rougny • Talk
A08	A disease agnostic pipeline for genome-scale metabolic modeling application in sample-specific drug combination ranking. Rui Afonso Tavares • Poster
A09	Modeling of NF-κB Pathway Rewiring Reveals Clarithromycin for IBD Philip Stegmaier • Talk
A10	EthoMap: A hybrid PD-AF knowledgebase of laboratory mouse behavior Rupert Overall • Talk
A11	Toward explainable Disease Maps: Identifying best practices for designing modeling-ready diagrams Pierre Klemmer • Poster
A12	AKOP: A Disease-Map Framework for Integrating Osteoarthritis Phenotypes, Molecular Networks and Transcriptomic Perturbations Suman Muralidharan • Talk
A13	Epigenomic Instability - Is Epigenetic Age Acceleration a Survival Indicator in Lung Cancer? Alexandra Baumann • Poster
A14	A multi-layer disease map to study post-COVID mechanisms Dries Heylen • Poster
A15	Specialized Pro-Resolving Lipid Mediators (SPMs): Literature Analysis and an LLM-Driven Knowledge Graph Vaishnavi Sirul Velaga • Talk + Poster
A16	Disease Maps as an Interface for Accessible Large-Scale Biomedical Data Analysis Matti van Welzen • Talk
A17	A Multi-Compartment Disease Map Approach to MASLD Matti van Welzen • Poster

A18	Network integration of liver-derived spatial transcriptomes from MASH patients Lukas Erkan Neumann • Poster
A19	A mechanistic pain signaling disease map enables pharmacophore-guided target prioritization and simulation of pain therapy response in osteoarthritis Shailendra Gupta • Talk
A20	NOSE-OE: A Molecular Signaling Map of the Olfactory Epithelium Shailendra Gupta • Talk
A21	Pathway Converter: A Desktop Workflow for Pathway Exploration Abrar Zahin • Talk
A22	Endometriosis-associated signaling pathways and potential for therapeutic development Geraldine Canny • Talk

Boolean network models recover the known biology of FSGS in podocytes and parietal epithelial cells

Raul Armando Duran De Alba, Olivia Lenoir, Pierre-Louis Tharaux, Clemence Reda

Institut de Biologie de l'École normale supérieure (IBENS) - PARCC INSERM, France

Focal segmental glomerulosclerosis (FSGS) is the scar lesion of some glomeruli. It is a rare disease, yet a devastating one, where about 50% of patients reach end-stage renal disease, and even after transplantation, FSGS recurs in 30 to 50% of cases. Often, two cell types are involved in FSGS: podocytes keep larger proteins out of the urine, while parietal epithelial cells (PECs) line Bowman's capsule and move into the tuft once podocytes are injured. Boolean networks are dynamical models that make gene regulation executable and support counterfactual perturbation predictions.

There is currently no executable model of either cell type, and no way to predict how they respond to perturbation. We report the first Boolean network models of FSGS, built automatically over thousands of genes from heterogeneous RNA-seq data in an open-source pipeline.

For each cell type, we refined gene-gene interactions from a meta-analysis of FSGS-related genes (Roointan et al., 2024), enriched with STRING annotations and filtered by co-expression across RNA-seq perturbation experiments: four in podocytes (VASN knockdown; LATS2, RARRES1, and GPR124 overexpression) and two cross-species in PECs (EGFR in mouse and TGFB1 in rat). We then inferred Boolean networks with NORDic, an inference tool we extended. We validated them by (1) scoring the recovery of reported STRING direct physical interactions, (2) testing the inferred modules for enrichment in known FSGS pathways, and (3) assessing prediction on 28 podocyte and 17 PEC-curated unseen experiments. We obtained one network per cell type, with an average of 1,000 genes each. Against the STRING physical interactions in the candidate network, they recover 48 to 51% precision and 22 to 30% recall (F1 0.30 to 0.38). Identified functional modules are related to FSGS. The podocyte network is built around growth factor signaling, with AKT1, FN1, and STAT3 as hubs, and carries a separate actin-cytoskeleton module holding ACTN4 and ARHGAP24. The PEC network centers instead on extracellular matrix and focal adhesion around FN1, CAV1, and COL4A1. Simulated perturbations reproduce literature-reported outcomes in 40% of podocyte and 24% of PEC cases.

These networks make FSGS mechanisms executable in each cell type and can be perturbed *in silico* to prioritize gene-level targets before experimental screening. They are also the first building blocks for modeling podocyte-PEC crosstalk and, eventually, the whole glomerulus.

A Hybrid computational Framework Integrating Boolean Regulatory Networks and Agent-Based modalities for modeling the growth of colorectal cancer organoids

Hala Bouzidi, Laurent Naudin, Philippe Castera, Guillaume Lefebvre
Dassault Systèmes BIOVIA, France

Colorectal cancer (CRC) is one of the most common causes of cancer-related deaths worldwide. CRC is driven by genetic mutations and dysregulated signaling pathways. Patient-derived organoids (PDOs) are powerful experimental models for studying CRC but are time-consuming and costly. Computational approaches provide complementary tools to investigate tumor biology while reducing experimental burden and offering mechanistic insights. In this context, Boolean and agent-based models are particularly relevant. We aim to develop a multiscale virtual organoid that integrates Boolean regulatory network modeling with agent-based spatial cell dynamics. The model will be calibrated using CRISPR and drug response data, building upon our proprietary ZhegAlCal tool (a calibration algorithm combining Zhegalkin polynomials and a weighted maxSAT solver). We have sourced 37 signaling pathways from the KEGG and Reactome databases and merged them all in a single large-scale naïve Boolean network. In parallel, we have improved our ZhegAlCal tool in order to efficiently handle highly connected nodes with up to 23 parent nodes. A computational pipeline was also developed to process raw CRISPR screen read count data into Gene Effect estimates using the Chronos algorithm. The inferred Gene Effects will be binarized and used to calibrate the Boolean network. The resulting Boolean model will further be integrated into a multi-agent framework, thus providing a predictive in silico platform for evaluating therapeutic strategies.

Establishing an omics-informed logic-based quantitative adverse outcome pathway framework using PFOA-induced toxicity as a case study

Luiz Ladeira, Arthur de Carvalho e Silva, Bernard Staumont, Liesbet Geris

GIGA Institute, University of Liège, Belgium

As toxicology transitions toward animal-free methods, there is a critical need to move beyond qualitative Adverse Outcome Pathways (AOPs) toward quantitative frameworks for probabilistic risk assessment. Perfluorooctanoic acid (PFOA) presents a challenge for risk assessment due to its accumulation and its capacity to trigger hepatotoxic outcomes via diverse molecular mechanisms. In this study, we aimed to develop a logic-based quantitative AOP (qAOP) model that translates transcriptomic signatures into probabilities of KE activation while considering cellular response variability. We built a molecular interaction network comprising 311 genes and their regulatory logic, targeting 14 liver-specific KEs adapted from a published AOP network for liver steatosis. The model was contextualized using a public in vitro-derived transcriptomics dataset of PFOA exposure (GSE144775). To account for intracellular heterogeneity, we generated a virtual population by applying gene-level variance to kinetic parameters. Continuous-time Markov simulations computed mean steady-state probabilities for all KEs across a 0-50 μM PFOA dose range at 24-hour acute and 10-day repeated exposures. The framework revealed the specific mode of action (MoA) of PFOA before the onset of cytotoxicity, yielding dose-response activities per KE that reflect phenotype behaviours over different perturbation profiles. Peroxisomal beta-oxidation increased from a 7% baseline to 84% at 20 μM (24h). Apoptosis showed increasing activities (from 5% at baseline 24h to 26% at 50 μM 10d, peaking at 49% at 20 μM 10d), reflecting cumulative toxicological impact over repeated exposure. Lipid accumulation showed higher response variation in both scenarios, yet an increasing trend, reflecting the experimental limitations of low biological replicates in the available data. This study establishes a prototype for transforming qualitative AOPs into probabilistic, omics-informed predictive tools with a granular, mechanistic view of individual KE activation. By incorporating kinetic variance, the model shifts from predicting isolated biological states to describing chemical-specific MoAs across acute and repeated regimes. This logic-based qAOP framework thus offers a scalable, mechanistically grounded approach, aligned with the transition toward quantitative systems toxicology.

Enhancing the performance and interpretability of epigenetic clocks

Tushar Patel, Steve Hoffmann, Alena van Bömmel

Leibniz Institute on Aging - Fritz Lipmann Institute (FLI), Germany

Epigenetic clocks based on DNA methylation (DNAm) accurately predict age, but their biological underpinnings remain unclear. One primary mechanism by which DNAm might influence gene regulation is by modulating transcription factor binding activity. This study investigates the regulatory potential of predictive CpGs in established epigenetic clocks. Our analysis reveals that generally most CpGs used by epigenetic clocks do not overlap known transcription factor binding sites (TFBS), indicating that clock accuracy is not primarily driven by changes in TF binding dynamics. However, analysis of CpGs within TFBSs identifies key transcription factors potentially involved in aging, including ZBED1, NFE2, and CEBPB, which are enriched for age-associated CpGs, while RELA, IKZF1, and STAT3 significantly protected against methylation changes. Leveraging TFBS-associated and age-correlated CpGs, combined with noise-stabilizing feature engineering steps, we developed an alternative TFMethyl Clock model that provides competitive predictions of chronological age [1]. Age-predictive CpGs selected by our model enrich for target genes involved in interleukin-1 β production and fatty-acid metabolism, while being enriched at TFBSs of NR2C2. Furthermore, approximately three-fourths of these target genes exhibit significant age-related changes, suggesting deeper insights into possible methylation-driven aging processes. Our findings demonstrate that incorporating regulatory information into epigenetic clocks may provide mechanistic insights into the aging process while improving the interpretability and predictive power.

Reference

1. Tushar Patel, Robert Schwarz, Konstantin Riege, Miri Varshavsky, Anne Richmond, Riccardo E Marioni, Hans A Kestler, Tommy Kaplan, Steve Hoffmann, Alena van Bömmel, Enhancing the performance and interpretability of epigenetic clocks, *Nucleic Acids Research*, Volume 54, Issue 13, 22 July 2026, gkag661, <https://doi.org/10.1093/nar/gkag661>

Identifying Key Events for Nitrogen Dioxide Effects: Toward an Adverse Outcome Pathway

Joshua Eugenio, Véronique de Bruijn, Jeroen Pennings, Arno Swart, Pim Wassenaar, Aurélie Carlier, Yvonne Staal

National Institute for Public Health and the Environment (RIVM), Netherlands

Background: Inflammation is a key process underlying acute and chronic respiratory effects. To determine whether exposure to airborne substances, such as nitrogen dioxide (NO₂), can lead to respiratory effects, it is important to elucidate the mechanisms involved in inflammation following inhalation. This study aims to develop an Adverse Outcome Pathway (AOP) by identifying and characterizing key biological events involved in inhalation-induced inflammation.

Methods: Scientific literature data were collected on the effects of NO₂ on the respiratory tract. Studies were included if they reported exposure concentrations (µg/m³, ppm, etc.) and duration, used human-relevant models, or (for animal/in vitro studies) assessed human relevance. Biological events and pathways were mapped using CellDesigner, which generates SBML (Systems Biology Markup Language) compliant network representations. The graphical notation provides visualization of the mechanisms involved while enabling identification of knowledge gaps and missing mechanistic links, guiding further literature searches to improve the network. Data on exposure levels and outcomes were systematically extracted and analyzed for temporal and dose-response trends using R.

Results: Data from 38 studies were collected. Of the 24 studies that included human data, 20 were in-vivo exposure studies (median cumulative exposure: 3 ppm-hour). Analysis shows that within 2h after exposure to relatively low concentrations of NO₂ (ppb-range) effects were mostly observed on components of the respiratory tract lining fluids (RTLFL), e.g., effects on small molecule antioxidants. Additionally, increases in markers associated with vasoconstriction, -dilation (e.g., Prostaglandin D₂, Thromboxane B₂) and chemoattractants (e.g., Interleukin-8, Leukotriene B₄) were observed. Within 24h, increasing numbers and induced activation of immune cells were observed (n=11 studies). We found that the interaction of NO₂ with immune cells is well supported in literature, while evidence for specific molecular and biochemical changes remains limited.

Conclusion: NO₂ quickly alters RTLFL components, followed by immunological changes. This evidence-based approach clarifies temporal and dose-dependent mechanisms of NO₂-induced inflammation. This will aid in defining the Molecular Initiating Events and Key Events, and how they qualitatively lead to the Adverse Outcome, ultimately enabling quantitative risk assessment for NO₂ and other inhaled toxicants.

LLM-Assisted Graph Extraction for Disease Map Construction

Joost Rehder, Suman Muralidharan, Vaishnavi Velaga, Falko Baustian

Systems Biology & Bioinformatics (SBI), University of Rostock, Germany

Manual construction and curation of disease maps and submaps is labor-intensive: a curator must read dozens of papers, decide which entities and interactions belong in the model, and ground each one in a reference database. We present a pipeline that turns a literature collection into an evidence-grounded, identifier-resolved graph, leaving the curator with a reviewable graph network rather than a blank canvas.

PubMed-indexed articles are processed by an LLM under a typed extraction schema, so that entities such as genes, proteins, and metabolites, together with their interactions, are returned as validated records and stored as nodes and edges in a graph database. The LLM used for extraction may be a public API or a locally hosted open-weight model. Each extracted entity is normalized against reference databases, among them NCBI Gene, UniProt and MeSH. Every entity node therefore carries a stable database identifier, which in turn drives entity resolution across publications, and every node retains provenance to its source sentence and PubMed ID.

The resulting graph supports two downstream uses. A GraphRAG-style agent with graph-query and full-text search tools answering natural-language questions and citing its answers back to the source passages. Since the nodes are typed and identifier-grounded, the representation aligns with CellDesigner entities and reactions, so transfer into an editable map is straightforward.

The momapy toolkit: Python tools for programmatic work with molecular maps

Adrien Rougny, Marek Ostaszewski, Venkata Satagopam

Luxembourg Centre for Systems Biomedicine (LCSB), University of Luxembourg

Molecular maps are graphical representations of the mechanisms underlying biological systems. They are a valuable tool for curating, exchanging, and understanding biological knowledge, and may serve as a backbone for data analysis and modeling. Molecular maps are supported by a rich software landscape, with tools to edit, render, convert, query or model them. These are however standalone applications, each implementing its own handling of maps, with no common programmatic representation. Here we introduce the momapy toolkit, a set of tools to work with molecular maps programmatically. At its core is momapy, a Python library that defines the common representation shared by all the tools of the toolkit. momapy separates the data model of a map from its graphical representation, and allows users to manipulate, render and compare maps. We also briefly introduce two other tools: momapy-kb, for storing and querying molecular maps in knowledge bases; and pd2af, a tool to transform process description maps into activity flow maps. The momapy toolkit introduces programmatic processing of molecular maps for streamlined standard verification, consistency checks, integration of different resources, and modeling workflows. The toolkit currently supports the SBGN and CellDesigner formats, two of the main standards to represent molecular maps graphically, and can be easily extended to support additional formats and functionalities. All tools are freely available at github.com/momapy under a GPLv3 license.

A disease agnostic pipeline for genome-scale metabolic modeling application in sample-specific drug combination ranking.

Rui Afonso Tavares; E. Stalidzans

Computational Systems Biology Group, Rīga Stradiņš University, Latvia

A computational pipeline for studying disease at the genome-scale level being developed combining various tools to propose personalized drug combination treatment according to sample-specific gene expression measurements. This pipeline allows the study of multiple diseases taking into account genome-scale mathematical modeling of metabolism that will serve as a valuable decision support system for medical doctors. First, a highly curated generic model of human metabolism (Human1) is used for integration of omics data. Following this, a simulation process is being developed in order to identify key aspects of the disease pathway that could be exploited for treatment. Depending on disease, key criteria will be assessed for appealing targets (for Cancer, reduction of cell biomass/proliferation and reduction of ATP production would be useful to evaluate). Assessment/scoring models for existing drugs for the disease of choice will be developed, and hypothetical combinations of drugs will be suggested and compared with existing treatments. Modeling targets and combinations will be simulated and tested in silico, compared to known treatment regimens and novel combination therapies will later be tested on samples in vitro.

Modeling of NF- κ B Pathway Rewiring Reveals Clarithromycin for IBD

Philip Stegmaier, Daria Stelmashenko, Kamilya Altynbekova, Alexander Kel

geneXplain GmbH, Germany

Keywords: disease-pathway modeling; NF- κ B dynamics; pathway rewiring; drug repurposing; inflammatory bowel disease

Motivation and Aim: Disease pathways are often represented as static interaction maps, although chronic inflammation emerges from state-dependent feedback, altered regulatory inputs, and changes in transcriptional output. In inflammatory bowel disease (IBD), persistent NF- κ B activity is a central disease mechanism. We aimed to reconstruct this pathway from extracellular signaling to nuclear gene regulation, model its transition from normal to inflammatory states, and identify drugs capable of disrupting the disease-specific network.

Methods: Differentially expressed genes were connected to their transcription-factor binding sites using TRANSFAC, GTRD and Hocomoco and then to upstream signaling regulators through TRANSPATH and Reactome. The resulting causal disease pathway linked receptors and signaling proteins to transcription factors and inflammatory gene programs. We modeled disease progression by dynamically modifying the structure of ordinary differential equations (ODEs) leading to formation of positive feedback loops that were fixing the disease state of the system. Such pathway rewiring was used to identify candidate targets followed by drug prioritization using PASS, which is a tool for QSAR-based chemoinformatics screening.

Results: We identified 159 NF- κ B pathway related candidate targets, including previously unrecognized regulators of sustained inflammatory signaling. Drug prioritization highlighted macrolide antibiotics, with clarithromycin as the top candidate predicted to perturb NF- κ B feedback dynamics. Experimental validation showed that clarithromycin suppressed NF- κ B p65 DNA binding in the small intestine of LPS-stimulated mice ($p = 0.002$). In DSS-induced colitis, it reduced weight loss ($p = 0.04$) and histological disease severity ($p = 0.004$). Murine and human intestinal enteroids further confirmed inhibition of NF- κ B nuclear translocation and signaling dynamics.

Conclusion: Modeling NF- κ B as a rewiring, state-dependent disease pathway—rather than a fixed diagram—connected omics-derived gene regulation with dynamic mechanisms and therapeutic intervention. The prediction and experimental validation of clarithromycin demonstrate how executable disease-pathway models can reveal causal control points and guide drug repurposing in IBD.

Funding: European Commission FP7 SysMedIBD project (No. 305564).

EthoMap: A hybrid PD-AF knowledgebase of laboratory mouse behavior

Rupert W Overall

Humboldt University of Berlin, Germany

The behavior of mice in a home cage setting can be monitored with modern technology to yield longitudinal ethological profiles. Perturbations in such profiles, due to age, disease, genetics or pharmacological intervention, allow detection of subtle imbalances in the underlying mechanistic pathways. High-throughput multidimensional behavioral phenotyping is becoming the norm and there exist enormous bodies of information on the physiological, neurological and molecular correlates of behavior. However formal representations of such information have been largely restricted to the molecular level. To bridge the chasm between genetics and ethology, I have initiated a map project to provide a human- and machine-readable knowledgebase that spans all biological levels between genetics, molecular interactions, physiological and neurological pathways and behavioral endophenotypes to causally link these with ethological observations. Because this map includes molecular detail, modeled in SBGN-PD, as well as higher-level biological activities, modeled in SBGN-AF, I have made use of the proposed SBGN Hybrid PD-AF language to link these scales. I also raise questions about the potential need for an SBGN extension to allow richer representation of non-molecular endophenotypes, which are currently under-supported in the AF language. My hope is that this project might be a template to encourage the wider use of clinically-relevant formal knowledgebases beyond the current molecular focus.

Toward explainable Disease Maps: Identifying best practices for designing modeling-ready diagrams

Pierre Klemmer, Samuel Pastva, David Šafránek, Marek Ostaszewski

Luxembourg Centre for Systems Biomedicine (LCSB), University of Luxembourg

Disease maps represent a unique interface between medicine, computational biology, and pharmacology owing to their scale and expert-curated content. Recent applications have shown that disease maps represent excellent scaffolds for Boolean modeling (Klemmer et al. 2026; Ostaszewski et al. 2021) that can approximate known disease mechanisms for in silico target identification.

The conversion of static disease maps to executable models requires a complex transition procedure implemented by tools like CasQ (Aghamiri et al. 2020). The success of this transition depends on the mechanistic precision of the input map, though many disease maps were drawn to document molecular processes and were not designed with simulation in mind. As such, these maps are formally valid but analytically impractical. Partially specified Boolean networks (Benes, Brim, Huvar, Pastva, and Šafránek 2023) represent sets of candidate models with incomplete update rules to allow the statement of regulatory uncertainties, enabling a pragmatic approach to modeling while being especially vulnerable to semantic inconsistencies in disease maps. For instance, the partially specified Boolean network derived from the mitophagy diagram of the Parkinson's disease map (Fujita et al. 2014) currently contains 115 variables of which 37 have partially specified update functions, resulting in over 22284 possible parametrisations of the model – an infeasible scale for exhaustive analysis that is in part due to curation decisions incompatible with modeling.

We present a computational workflow to close the gap between maps and models. Starting from an existing diagram, we diagnose which curation decisions lead to modeling challenges at three levels: topological properties of the diagram (e.g. ambiguous reaction types, phenotypes as reaction inputs), properties of the resulting partially specified Boolean network, and the model's input-output behavior against a suite of expected perturbation responses. Every diagnostic is versioned and links a map pattern to a modeling consequence to establish a set of rules and best practices for curators. Our pipeline implements modern network modeling tools including momapy (Rougny, Ostaszewski, and Sagatopam 2026), AEON.py (Benes, Brim, Huvar, Pastva, Šafránek, and Smijakova 2022), and Newt (Balci et al. 2021) to produce a modeling-friendly version of the mitophagy pathway for future applications in Boolean network inference, analysis, and logic reasoning.

AKOP: A Disease-Map Framework for Integrating Osteoarthritis Phenotypes, Molecular Networks and Transcriptomic Perturbations

Suman Muralidharan, Paulina Christen, Vaishnavi S. Velaga, Matti van Welzen, Konstantin Cesnulevicius, David Lescheid, Myron Schultz, Olaf Wolkenhauer, Alberto Migliore, Yves Henrotin, Shailendra Gupta

Systems Biology & Bioinformatics (SBI), University of Rostock, Germany

Knee osteoarthritis (KOA) is a heterogeneous disease involving interconnected molecular processes across cartilage, bone, synovium and other joint tissues. Representing these mechanisms in a structured and computable form is essential for integrating heterogeneous datasets and enabling systems-level analysis. We developed the Atlas of Knee Osteoarthritis and Pain Signaling (AKOP) as a modular disease-map resource connecting molecular mechanisms of KOA with structural phenotypes and experimental data.

AKOP currently comprises 39 interconnected mechanistic submaps integrating molecular processes across joint tissues, supported by evidence curated from hundreds of publications. The map organizes molecular entities and their interactions into interconnected mechanistic modules, providing a structured representation of disease-associated processes. Biological entities and their interactions are systematically curated and cross-referenced with established databases and ontologies. This modular network architecture enables molecular mechanisms to be explored across submaps and provides a basis for computational analysis of the disease map.

To extend AKOP from knowledge representation toward computational modeling, signal propagation analysis is being investigated to quantify how molecular perturbations influence downstream components of the disease network. Perturbation of one or multiple nodes can be propagated through activating and inhibitory interactions to estimate changes in downstream molecular activity. This provides a framework for comparing the network-level consequences of different perturbations and identifying processes that may be jointly affected across interconnected AKOP modules. In parallel, experimental transcriptomic datasets can be mapped onto AKOP to relate observed molecular changes to the curated disease mechanisms.

AKOP provides a structured interface between curated mechanistic knowledge and experimental data. Integrating disease-map representation with computational perturbation approaches could enable systematic exploration of how local molecular perturbations propagate across interconnected KOA mechanisms, supporting mechanistic interpretation and hypothesis generation.

Epigenomic Instability - Is Epigenetic Age Acceleration a Survival Indicator in Lung Cancer?

Alexandra Baumann, Michael Seifert, Zholdas Buribayev, Olaf Wolkenhauer, Markus Wolfien
Systems Biology & Bioinformatics (SBI), University of Rostock, Germany

Cancer genomes are characterized by a considerable amount of modifications on various multi-omics levels, often caused by genomic and epigenomic instability. This cancer hallmark represents, e.g., epigenetic alterations in DNA methylation, histone remodeling, and non-coding RNA regulation. Understanding these patterns can give insights into tumor evolution, heterogeneity, and therapeutic resistance. However, it requires both robust data infrastructure and mechanistic insight.

To address the computational challenge of handling large-scale cancer genomic data, we developed TCGADownloadHelper. This streamlined pipeline simplifies data retrieval from The Cancer Genome Atlas (TCGA) via the GDC portal. Thus, transparent preprocessing of multimodal, patient-linked datasets is possible. This tool formed the foundation for our downstream analyses of DNA methylation data across TCGA lung cancer cohorts.

Building on these resources, we investigated epigenetic age acceleration (EAA) across the lung cancer cohorts TCGA-LUAD and TCGA-LUSC. Illumina 450K DNA methylation data was obtained from the GDC portal via the TCGADownloadHelper. The deviation between biological (epigenetic) and chronological age was estimated from DNA methylation patterns using established epigenetic clocks (Horvath, Zhang2019,...). We demonstrate that approximately two thirds of tumors exhibited accelerated epigenetic aging, while one third showed a "rejuvenating" effect. Interestingly, patients with rejuvenating EAA showed significantly lower overall survival in Kaplan-Meier analyses. EAA varied by sex, tumor stage, and cancer type, highlighting the importance of patient-specific factors in the epigenetic age. Together, these findings introduce epigenetic aging signatures as a promising stratification biomarker and motivate joined mechanistic studies on genomic and epigenomic instability to characterize actionable targets in precision oncology.

A multi-layer disease map to study post-COVID mechanisms

Dries Heylen, Michelle Moerel, Martina Summer-Kutmon

Maastricht Centre for Systems Biology and Bioinformatics (MaCSBio), Maastricht University, Netherlands

A large group of post-COVID patients experience persistent symptoms including dysautonomia, post-exertional malaise, fatigue, and cognitive impairment, pointing toward ongoing biological disruption. Accumulating evidence implicates neuroinflammation, blood-brain barrier (BBB) dysfunction, and dysregulation of the autonomic nervous system (ANS), hypothalamic-pituitary-adrenal (HPA) axis, and immune system. Studying how these processes interact requires an integrative, multi-scale approach.

Drawing on data from three Dutch post-COVID cohorts (COVFATI, DECI-LOCO, IMPACO), we present the first version of our post-COVID disease map. What distinguishes this map within the disease map field is its explicit organization across interconnected layers: a phenotype layer capturing clinical symptoms and subgroups (e.g. symptom co-occurrence network), biological system layers capturing dysfunction across key physiological systems implicated in post-COVID pathology (e.g. brain networks), and molecular layers detailing the underlying pathways and cellular interactions. A top-level anatomical overview links to submaps detailing specific disease mechanisms, and clickable elements connect directly across layers and to computational models. This will allow users to trace how molecular-level perturbations potentially propagate through biological systems to a phenotype.

A key feature of the map is its support for experimental data annotation and overlay. Molecular, imaging, and clinical data can be directly integrated onto the map, merging mechanistic insight with patient data. This multi-layer structure facilitates pathway and network-based analyses, supporting biomarker discovery. Overlaying patient subgroup data (e.g., by sex, age, symptom severity) onto shared pathway structures further enables stratification analyses, revealing whether distinct clinical presentations map onto distinct underlying mechanisms.

The disease map was constructed by integrating cohort data with current literature following community-driven standards for integrated disease maps, capturing our understanding of the interplay between neuroinflammation, BBB disruption, and neuroendocrine and immune dysregulation. The map is under active development: current work focuses on expanding pathway coverage and refining cross-layer connectivity, and it will remain a living resource, iteratively updated as new knowledge and experimental data become available.

Specialized Pro-Resolving Lipid Mediators (SPMs): Literature Analysis and an LLM-Driven Knowledge Graph

Vaishnavi Sirul Velaga, Suman Muralidharan, Matti van Welzen, Konstantin Cesnulevicius, David Lescheid, Myron Schultz, Olaf Wolkenhauer, Shailendra Gupta

Systems Biology & Bioinformatics (SBI), University of Rostock, Germany

Specialized Pro-Resolving Mediators (SPMs) are bioactive lipid mediators that actively regulate the resolution of inflammation and promote the restoration of tissue homeostasis. They comprise several families, including resolvins, protectins, maresins, lipoxins, and Cys-SPMs, and influence a broad range of molecular targets and signaling pathways. However, knowledge about SPM biosynthesis, receptors, molecular targets and downstream biological effects is scattered across the scientific literature, making it difficult to obtain an integrated view of SPM biology. We therefore developed SPIN (Specialized Pro-resolving mediator Interactome), an AI-assisted framework to automatically extract, organize and explore SPM-related molecular interactions from biomedical literature.

Publications related to individual SPMs were retrieved from PubMed. A total of 2,348 publications were retrieved. Biomedical entities, including genes, proteins, chemicals and diseases were identified using SciSpacy Named Entity Recognition models and mapped to relevant standardized biomedical databases, such as UniProt and PubChem. BioMistral-7B, a biomedical Large Language Model (LLM), was employed to extract biologically meaningful interaction triplets representing regulatory relationships, biosynthetic processes, and disease associations. Each extracted interaction was linked to its supporting textual evidence and corresponding PubMed identifier. Evidence similarity and LLM confidence scores were combined to assess the reliability of the extracted interactions. The resulting interactions were integrated into a knowledge graph connecting SPMs with molecular and biological entities while maintaining links to the supporting literature. An interactive web interface enables visualization of SPM-specific networks, exploration of molecular interactions, confidence-based filtering, and natural-language querying of the knowledge graph.

SPIN transforms unstructured SPM literature into structured and traceable interaction networks. It allows researchers to explore and compare molecular targets and pathways across different SPMs while keeping each interaction connected to its supporting evidence. Overall, SPIN provides an interactive web based resource for exploring SPM biology and supporting a systems-level understanding of inflammation resolution.

Disease Maps as an Interface for Accessible Large-Scale Biomedical Data Analysis

Matti van Welzen, Shailendra Gupta, Olaf Wolkenhauer

Systems Biology & Bioinformatics (SBI), University of Rostock, Germany

The Research Training Group SYLOBIO investigates local and systemic inflammatory responses to biomaterials used in joint replacements and the treatment of skin lesions. Across its interdisciplinary projects, SYLOBIO generates heterogeneous experimental, molecular, clinical, and epidemiological data that need to be analyzed across biological scales and in the context of existing mechanistic knowledge. This raises the challenge of how such diverse datasets can be integrated and analyzed in a way that is biologically interpretable, and at the same time accessible to researchers across an interdisciplinary group. Here, we present LUMINAR, a plugin for the MINERVA platform that extends disease maps from resources for knowledge representation and visualization into interactive environments for data exploration and analysis. Experimental molecular datasets can be uploaded or retrieved from FAIRDOMHub, processed, mapped to disease-map entities, and analyzed directly in the context of the molecular network. LUMINAR combines data integration with network-based approaches, including two-dimensional enrichment analysis (2DEA), to translate molecular measurements into network-level information. Downstream and upstream analyses connect observed molecular changes with affected biological processes and regulatory mechanisms. These results can be explored across samples, experimental conditions, time points, or datasets while retaining the connection between analytical results and the underlying disease-map network. A central methodological component of LUMINAR is the use of large language models as a layer between researchers and computational methods. Rather than the LLM performing the analysis, natural-language questions are translated into workflows of data retrieval, network analysis, statistical analysis, and visualization operations. The same approach also enables conversational exploration of map structure, molecular interactions, paths, and hypothetical perturbations. Using SYLOBIO as the motivating application, we demonstrate how disease maps, network-based analysis, research data infrastructure, and LLM-guided tool use can be combined into a common analysis environment. This approach aims to lower the technical barriers to large-scale biomedical data analysis while maintaining transparent computational workflows and a direct connection between experimental observations and mechanistic biological knowledge.

A Multi-Compartment Disease Map Approach to MASLD

Matti van Welzen, Ronja Lea Jennifer Müller, Ali Canbay, Andreas Geier, Jörn M Schattenberg, Konstantin Cesnulevicius, Myron Schultz, David Lescheid, Olaf Wolkenhauer, Shailendra Gupta
Systems Biology & Bioinformatics (SBI), University of Rostock, Germany

Metabolic dysfunction-associated steatotic liver disease (MASLD), formerly known as non-alcoholic fatty liver disease (NAFLD), has a complex pathogenesis that requires an in-depth understanding of diagnostic and prognostic molecular patterns to develop effective clinical treatment strategies. Due to the heterogeneity of the liver, approaches that focus on small-scale models of a few signaling pathways might inadequately represent the spatial aspects of disease progression. To this end, we created a disease map and collected experimentally validated molecular interactions to study macro-scale heterogeneity and disease progression. We developed a multicompartmental agent-based model (ABM) of autonomous but interacting Boolean networks representing heterogeneous liver compartments and cell types. The stochastic distribution of nutrient supply and metabolic activities between compartments allows the simulation of variations in liver metabolism under different conditions. We present the model as an interactive web-based platform comprising over 30 interconnected subnetworks following SBML standards. Network models of digestive processes enable the study of disease progression following changes in nutritional status. Finally, integrating models of the immune response, such as from the Atlas of Inflammation Resolution (AIR) [1], establishes links between metabolic changes in liver compartments and inflammation, bridging the gap to metabolic dysfunction-associated steatohepatitis (MASH). The disease map allows researchers to perform *in silico* simulations of MASLD development, defining parameters such as diet and drug interactions. The impact of changing conditions on MASLD progression can be visualized through simulations, providing a better understanding of the disease. Our resource supports the simulation of clinical interventions, potentially facilitating the development of improved treatment strategies.

Network integration of liver-derived spatial transcriptomes from MASH patients

Lukas Erkan Neumann; Matti van Welzen; Shailendra Gupta

Systems Biology & Bioinformatics (SBI), University of Rostock, Germany

Metabolic dysfunction-associated steatohepatitis (MASH) is characterized by chronic hepatic inflammation and can progress to fibrosis through complex interactions between hepatic, stromal, and immune cell populations. Spatial transcriptomics enables the investigation of these processes while preserving the spatial organization of the tissue. We characterized fibrosis-associated changes in cellular composition, spatial organization, and intercellular communication in human MASH liver tissue using Xenium spatial transcriptomics.

Spatial transcriptomic data from 16 patient samples with MASH with or without fibrosis were analyzed using a computational pipeline integrating cell segmentation, cell-type annotation, compositional and spatial analyses, and ligand–receptor inference. Baysor-based segmentation and marker-guided cell-type annotation were evaluated to improve transcript assignment and enable downstream single-cell analyses.

Overall cell-type composition was broadly similar between conditions, although fibrosis-associated trends were observed among hepatic stellate cell and macrophage populations. Spatial analyses revealed distinct organization of activated hepatic stellate cells into fibrotic niches enriched in infiltrating monocytes and macrophage populations, including C-LAMs, LAMs, and monocyte-derived Kupffer cells. Ligand–receptor analysis further indicated differences in potential intercellular communication. While MASH was predominantly associated with inflammatory and regenerative signaling, MASH fibrosis showed a more coherent chemokine-dominated pattern involving CCL2, CCL3, and CCL3L1 signaling through CCR3 and CCR5, with prominent contributions from macrophage and hepatic stellate cell populations.

Together, these findings suggest that fibrosis in MASH is associated not only with changes in cellular abundance but particularly with spatial reorganization and altered myeloid–stromal communication. The identified cellular niches and signaling interactions provide hypotheses for fibrosis-associated mechanisms that require further functional validation.

A mechanistic pain signaling disease map enables pharmacophore-guided target prioritization and simulation of pain therapy response in osteoarthritis

Paulina Christen, Vaishnavi S. Velaga, Anneke Lemken, Soorya Jankiraman, Suman Muralidharan, Matti van Welzen, Konstantin Cesnulevicius, David Lescheid, Myron Schultz, Olaf Wolkenhauer, Alberto Migliore, Yues Henrotin, **Shailendra Gupta**

Systems Biology & Bioinformatics (SBI), University of Rostock, Germany

Pain in osteoarthritis (OA), including knee OA and inflammatory OA phenotypes overlapping with rheumatoid arthritis (RA), results from complex interactions between inflammatory mediators, nociceptor signaling, and neuroimmune sensitisation. Clinical response to pain-relieving therapies is highly variable, underscoring the need for mechanistic frameworks that link drug modes of action to patient-reported pain outcomes. To develop a computable pain signaling disease map for osteoarthritis and to demonstrate its application for pharmacophore-guided target identification, network-based simulation of drug modes of action, and stratification of pain therapy responders. Seven core pain-related signaling pathways relevant to knee osteoarthritis were identified. PubMed abstracts were retrieved using the Biopython package and analyzed with a large language model (Mistral AI) to extract molecular interaction triplets (Node-1, Interaction, Node-2). Curated interactions were integrated into pathway-specific knowledge graphs and merged into a unified pain signaling disease map using CellDesigner. Network topology analyses were applied to identify mechanistic hub nodes based on centrality, pathway crosstalk, and regulatory influence. In a representative case study, pharmacophore modeling was applied to selected analgesic drugs to identify putative human protein targets with available structural data (PDB). Identified targets were mapped onto the pain signaling network and subjected to in silico perturbation and signal-propagation analyses. Patient-reported pain scores from the Osteoarthritis Initiative (OAI) cohort were used to anchor network responses to clinical pain phenotypes. The pain signaling disease map integrates seven pain-related pathways into a unified mechanistic framework comprising inflammatory mediators, prostaglandin and leukotriene signaling, ion channels, neurotrophic factors, and central sensitisation modules. Network topology analysis identified a set of highly connected hub nodes linking inflammatory and neuronal pain pathways, highlighting regulatory bottlenecks in OA pain signaling. Pharmacophore-guided target identification recovered several established pain- and inflammation-relevant proteins, including peroxisome proliferator-activated receptors (PPAR γ and PPAR δ), carbonic anhydrase II, integrins, and neuroinflammatory regulators, supporting the biological plausibility of the approach. Network-based perturbation simulations demonstrated that different target classes induce distinct pathway-level responses, with PPAR-associated perturbations preferentially modulating inflammatory signaling, while other targets showed stronger effects on nociceptor and sensitization modules. Integration of patient-reported pain scores from the OAI cohort enabled phenotype-anchored mapping of pain severity onto the network and supported stratification of patients into mechanistically distinct responder and non-responder profiles based on dominant pathway activity patterns. We present a systems-level pain signaling disease map that supports pharmacophore-guided target prioritization and mechanistic simulation of drug modes of action in osteoarthritis. By integrating causal network structure, drug–target relationships, and clinical pain data, this framework provides a foundation for precision pain management and patient stratification in OA.

NOSE-OE: A Molecular Signaling Map of the Olfactory Epithelium

Shailendra Gupta, Federica Genovese, Krishna P Singh, Alessandro Nicoli, Francesco Ferri, Malgorzata M. Kogut-Günthel, Dominique Fastus, Suchi Smita Gupta, Matti van Welzen, Olaf Volkenhauer, Antonella Di Pizio

Systems Biology & Bioinformatics (SBI), University of Rostock, Germany

The olfactory epithelium (OE) is a dynamic and complex sensory tissue that is constantly challenged by external factors (such as pathogens, bacteria) and internal factors (such as neurodegeneration, inflammation, aging). Understanding the intricate cell-to-cell communication and regulatory networks that underline critical OE processes, including continuous cell regeneration, differentiation of olfactory sensory neurons (OSNs), and maintenance of supporting structures, remains a major challenge. To address this, we developed NOSE-OE (Network of Signaling Events in the Olfactory Epithelium), an empirical, open-access resource that curates, formalizes, and unifies fragmented mechanistic data from over 300 literature sources. We have structured this knowledge base into a standardized, computer-readable molecular signaling map containing over 500 molecular elements and 1,000 verified interactions, implemented natively within the MINERVA platform framework (https://air.elixir-luxembourg.org/minerva/index.html?id=NOSE_OE_v01). This platform enables intuitive exploration of curated knowledge and supports hypothesis-driven interrogation of OE signaling networks, thereby providing an analysis-ready resource to foster systems-level discoveries into olfactory epithelial biology in health and disease.

NOSE-OE: A Molecular Signaling Map of the Olfactory Epithelium

Abrar Zahin, Shailendra Gupta

Systems Biology & Bioinformatics (SBI), University of Rostock, Germany

The olfactory epithelium (OE) is a dynamic and complex sensory tissue that is constantly challenged by external factors (such as pathogens, bacteria) and internal factors (such as neurodegeneration, inflammation, aging). Understanding the intricate cell-to-cell communication and regulatory networks that underline critical OE processes, including continuous cell regeneration, differentiation of olfactory sensory neurons (OSNs), and maintenance of supporting structures, remains a major challenge. To address this, we developed NOSE-OE (Network of Signaling Events in the Olfactory Epithelium), an empirical, open-access resource that curates, formalizes, and unifies fragmented mechanistic data from over 300 literature sources. We have structured this knowledge base into a standardized, computer-readable molecular signaling map containing over 500 molecular elements and 1,000 verified interactions, implemented natively within the MINERVA platform framework (https://air.elixir-luxembourg.org/minerva/index.html?id=NOSE_OE_v01). This platform enables intuitive exploration of curated knowledge and supports hypothesis-driven interrogation of OE signaling networks, thereby providing an analysis-ready resource to foster systems-level discoveries into olfactory epithelial biology in health and disease.

NOSE-OE: A Molecular Signaling Map of the Olfactory Epithelium

Geraldine Canny

South East Technological University (SETU), Ireland

Endometriosis, a debilitating, whole body disease is estimated to affect 1 in 10 women of reproductive age and is a major cause of chronic pain and infertility. Patients may undergo several surgeries and current pharmacological treatments are associated with undesirable side effects.

Endometriosis is characterised by a persistent inflammatory microenvironment within the peritoneal cavity and elsewhere, driven in part by aberrant immune cell function. Steroid hormones are also thought to play a role in the pathology. Several studies have shown augmented production of pro-inflammatory cytokines and angiogenic growth factors.

Certain experimental observations are consistent with a failure of endogenous inflammation resolution. Under physiological conditions, resolution of inflammation is an active process orchestrated by specialized pro-resolving mediators (SPMs), including lipoxins, resolvins, protectins, and maresins. These lipid mediators limit further leukocyte recruitment, promote macrophage-mediated efferocytosis, facilitate tissue repair, and restore homeostasis without causing immunosuppression. Although direct evidence for impaired SPM biosynthesis in human endometriosis remains limited, the disease exhibits many hallmarks of defective resolution biology, including persistent macrophage activation, ongoing cytokine production, excessive angiogenesis, and inadequate clearance of ectopic tissue. We and others have demonstrated that exogenous Lipoxin A₄ suppresses inflammatory and estrogen-mediated signalling, reduces prostaglandin production and attenuates lesion development in animal disease models, suggesting that augmentation of pro-resolving pathways may represent a novel therapeutic strategy for endometriosis. Estrogen receptors bind and/or are activated by structurally diverse molecules. A body of research also suggests that certain environmental toxicants are also capable of acting as endocrine disruptors, potentially affecting the development and progression of endometriosis in certain women. Thus, it is becoming increasingly important to examine how various molecules, capable of acting as steroidal signals within the reproductive tract, may impact immune function related to either promotion or inhibition of this disease. This presentation will provide an overview of associated signaling pathways, giving an insight into the complexity of immuno-endocrine crosstalk.